VPD SURVEILLANCE UNIT

GOVERNMENT OF TAMIL NADU / 2026

VELLORE DISTRICT

Suspected Measles Case Analysis

January – July 2026 | 73 Suspected Cases | 35 PHCs / UPHCs

HSC-LEVEL SURVEILLANCE REPORT

Executive Summary

Key epidemiological indicators — Vellore District, Jan – Jul 2026

73
Total Suspected Cases
Jan – Jul 2026

35
PHCs / UPHCs Affected
Across Vellore District

38%
Cases Under Age 1
28 of 73 cases

62%
Completely Immunised
45 of 73 cases

Peak month: March 2026 — 17 cases (23.3% of total)
Primary reporting source: CMC Vellore (42.5%) followed by GVMCH Vellore (21.9%)

Surveillance Background

VPD surveillance in Vellore District — data source and scope

CONTEXT

Measles-Rubella Surveillance

Suspected measles cases reported to the State MR Surveillance Unit under the Vaccine Preventable Disease (VPD) programme, Tamil Nadu.

Case Investigation Form (CIF)

Each case recorded using standardised CIF capturing demographic, clinical, immunization, and epidemiological details.

Unique EPID Identifier

Each case assigned a 15-character alphanumeric EPID code (format: TN-VLR-26-XXX) for tracking and deduplication.

DATA SCOPE

Period
January 2026 – July 2026 (7 months)

Coverage
35 PHC / UPHC areas across Vellore District

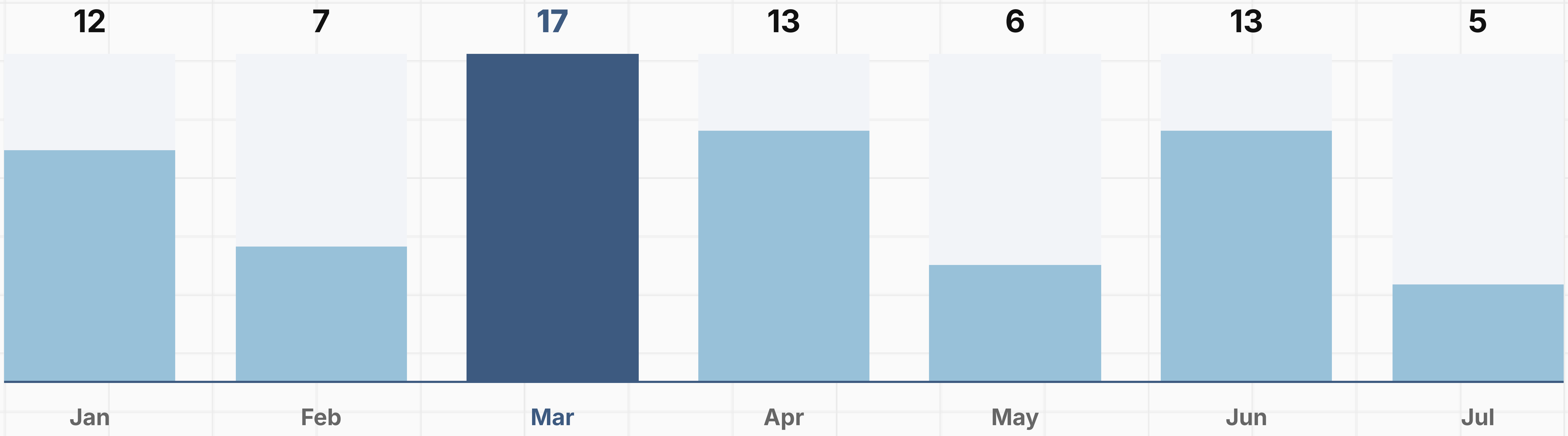
Total Cases
73 suspected MR cases (EPID: TN-VLR-26-001 to TN-VLR-26-110)

Reporting Sources
CMC Vellore, GVMCH, GPH, CHAD, and local PHCs / UPHCs

Coverage Metrics
MR-I and MR-II performance data included per HSC sector

Monthly Case Distribution

Suspected measles cases by month — January to July 2026 (Total: 73)

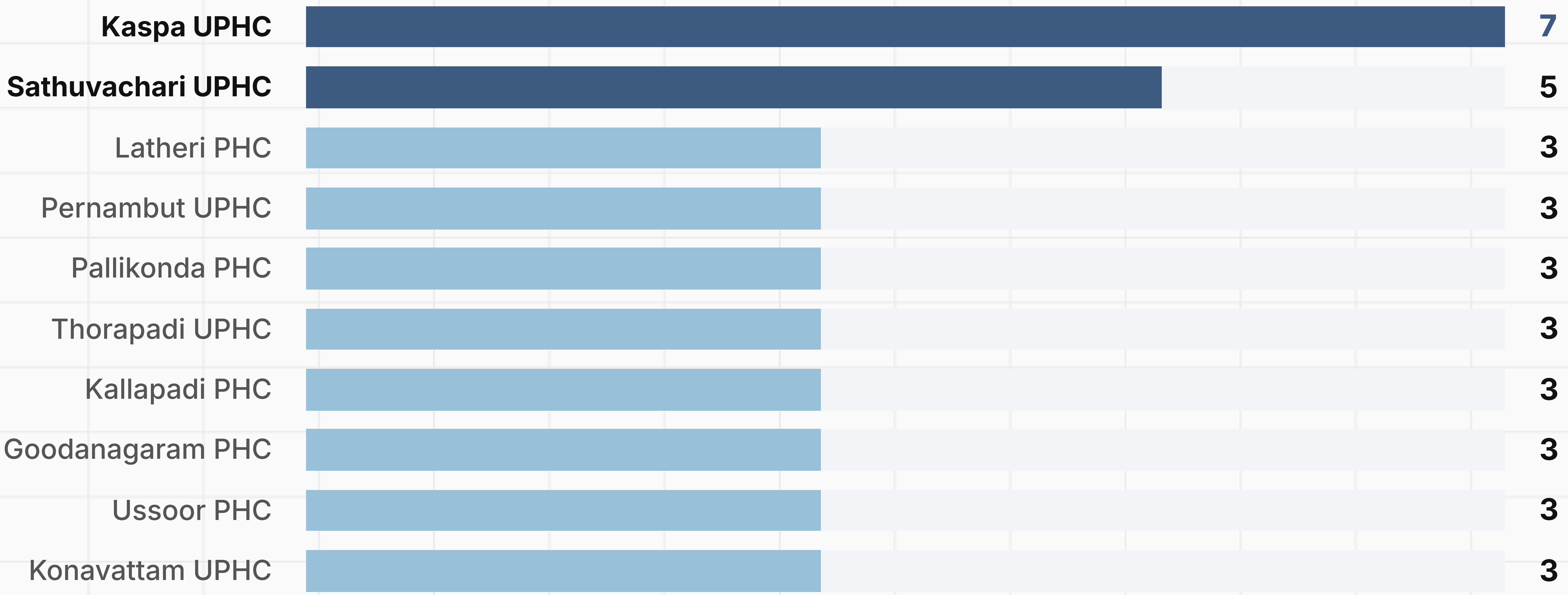


March 2026 recorded the highest burden — 17 cases, representing 23.3% of all cases

A second rise in June (13 cases) suggests continued community transmission through mid-2026

PHC-wise Case Distribution

Top 10 PHC / UPHC areas by suspected case load — Jan to Jul 2026



25 additional PHCs / UPHCs reported 1–2 cases each. Total PHCs affected: 35

Gender Distribution

Sex-wise breakdown of 73 suspected measles cases

MALE

41

56.2%

of total suspected cases

FEMALE

32

43.8%

of total suspected cases

SEX RATIO

128 : 100

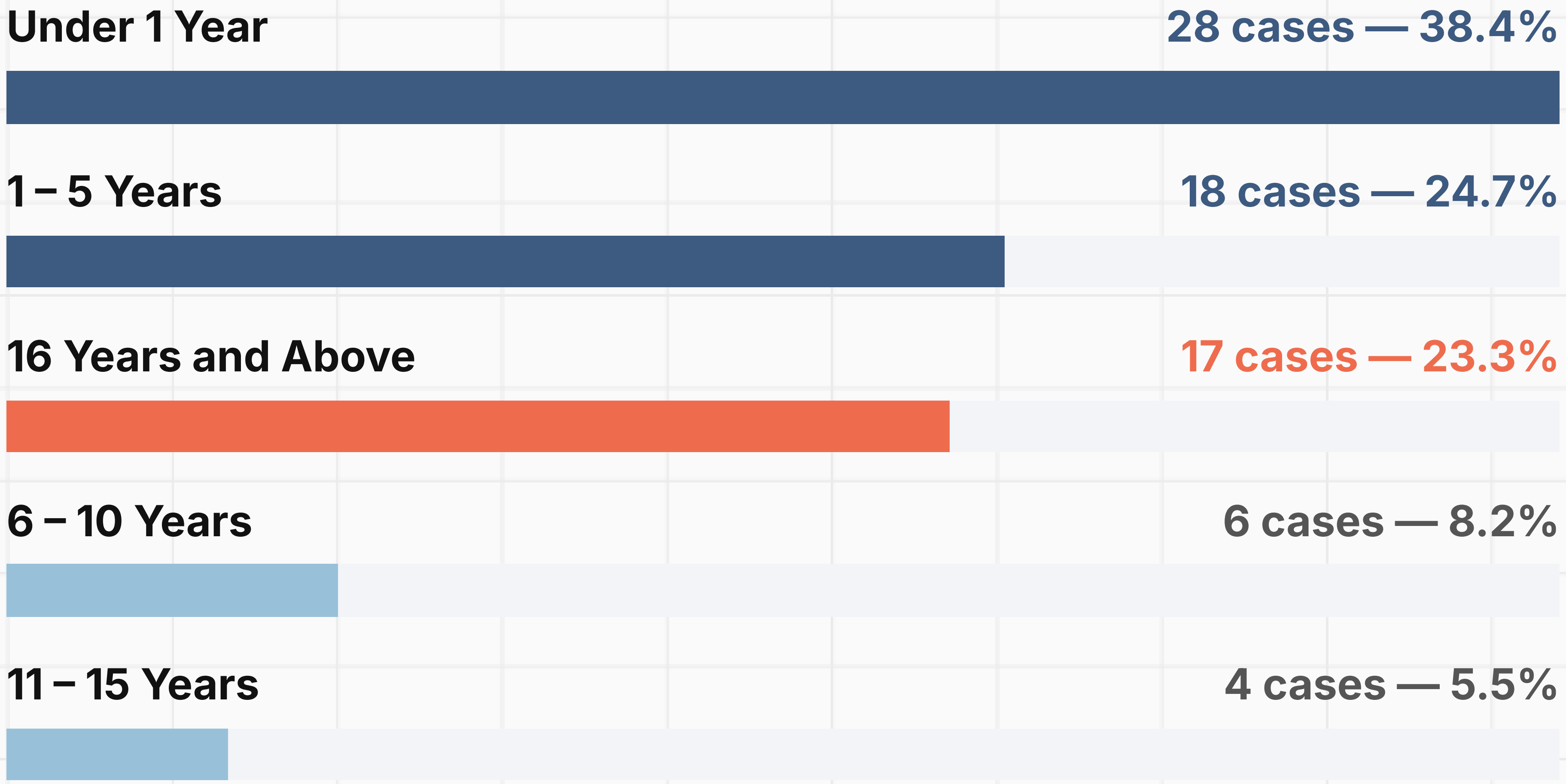
Male to Female ratio among cases

Observation

Male cases outnumber female by 28%. Both sexes affected across all age groups, suggesting equal exposure in community settings.

Age Group Analysis

Distribution of 73 suspected cases across five age categories



KEY FINDING

63%

of cases are children under 5 years of age (46 of 73 cases)

ALERT

17 adult cases (age 16+) detected — includes cases aged 27–50 years, indicating susceptibility gaps in older unvaccinated cohorts.

Immunization Status

MR vaccine status at time of case detection — 73 suspected cases

45

Completely Immunised

61.6%

Received MR dose(s) as per age schedule — yet developed suspected measles

24

Not Known

32.9%

Immunization history unavailable — highlights documentation and tracking gaps

4

Partially Immunised

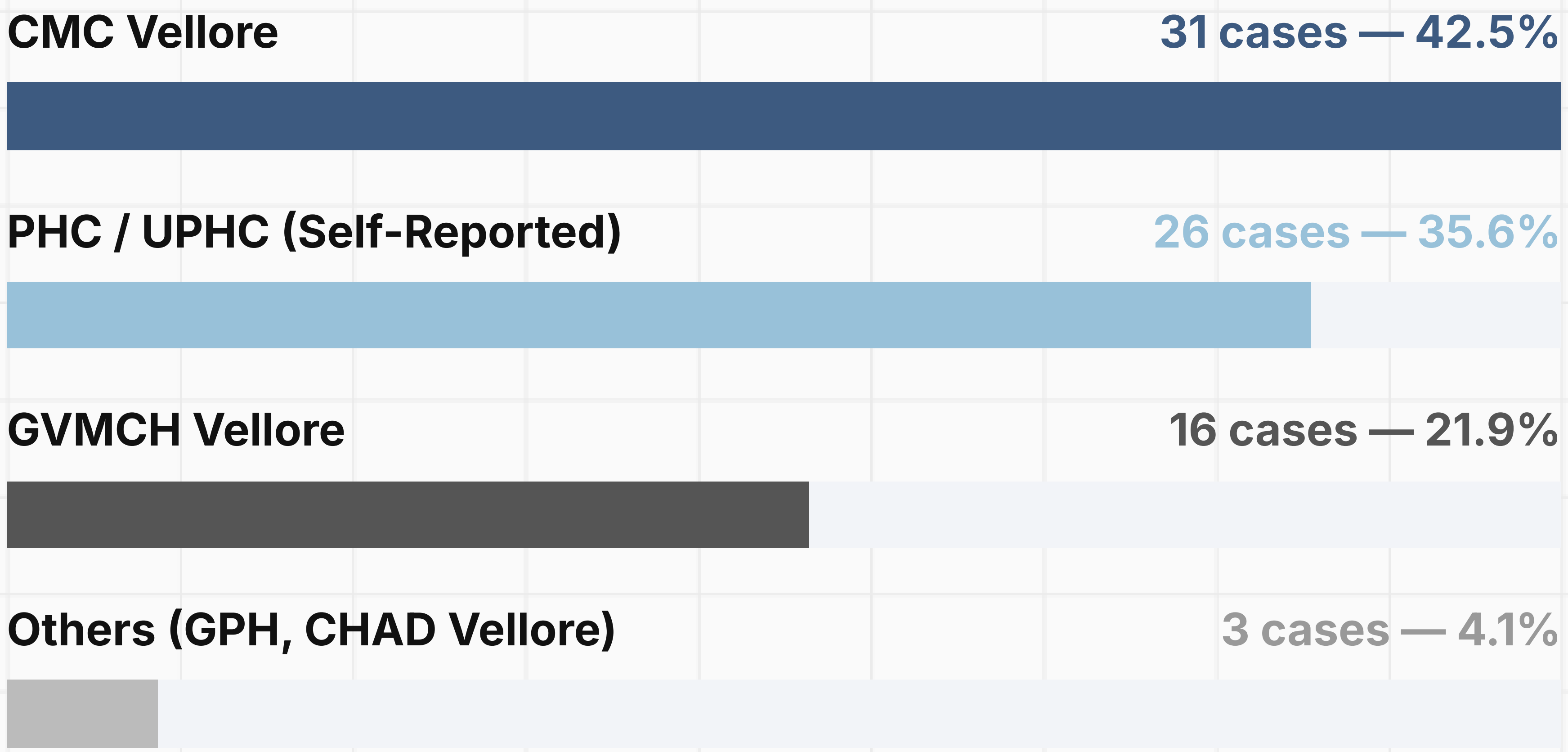
5.5%

Received only one MR dose — incomplete protection against measles

Concern: 61.6% of cases had complete immunization — requires investigation of cold chain, vaccine efficacy, or documentation accuracy

Source of Reporting

Facilities from which suspected cases were identified and reported



TERTIARY CENTRE BURDEN

64%

of cases identified through CMC and GVMCH — tertiary referral hospitals

IMPLICATION

High tertiary-centre reporting suggests possible under-detection at primary health level. Strengthening PHC-level surveillance is critical.

MR-I Immunization Coverage

MR Dose-1 performance (%) by HSC sector — representative sample

HSC / SECTOR	PHC / UPHC	MR-I %	STATUS
Ammundi	Thiruvalam PHC	120%	Over Target
T.V.K.Nagar	Pernambut UPHC	104%	Over Target
K.V.Kuppam	K.V.Kuppam PHC	112%	Over Target
Serpadi	Odugathur PHC	96%	Near Target
Nadupet	Nadupet UPHC	96%	Near Target
Muslimpur	Melpatti PHC	83%	Below Target
Velapadi	Lakshmipuram UPHC	82%	Below Target
Kagithapattarai	Sathuvachari UPHC	77%	Below Target

Target: 100% coverage for MR-I. Values above 100% may reflect migrant or out-of-area child vaccinations.

MR-II Immunization Coverage

MR Dose-2 performance (%) by HSC sector — representative sample

HSC / SECTOR	PHC / UPHC	MR-II %	STATUS
Ammundi	Thiruvalam PHC	117%	Over Target
Kagithapattarai	Sathuvachari UPHC	125%	Over Target
T.V.K.Nagar	Pernambut UPHC	93%	Near Target
Serpadi	Odugathur PHC	98%	Near Target
Muslimpur	Melpatti PHC	79%	Below Target
R.N.Palayam	Kaspa UPHC	73%	Below Target
Velapadi	Lakshmipuram UPHC	60%	Below Target
Salavanpalayam	Ambethkar UPHC	36%	Critical Gap

MR-II gaps are often wider than MR-I — second dose follow-up requires targeted outreach for missed beneficiaries.

MR Dropout Analysis

MR-I to MR-II dropout rate by HSC — positive value = children missed for second dose

HSC / SECTOR	PHC / UPHC	MR-I DROPOUT	MR-II DROPOUT
R.N.Palayam	Kaspa UPHC	-2	+8
Vallalar	Alamelumangapuram UPHC	-4	+6
Muslimpur	Melpatti PHC	+4	+5
Velapadi	Lakshmipuram UPHC	+2	+3
Chinnaallapuram	Kaspa UPHC	-5	+6
Arasamarapet	Ambethkar UPHC	-3	+1

HOW TO READ

- Positive (+): children not returning for 2nd dose
- Negative (-): more 2nd doses than 1st (catch-up)

Concern Areas

R.N.Palayam (+8) and Vallalar (+6) show the highest MR-II dropout — these sectors require targeted follow-up to complete the two-dose schedule.

High Burden PHC Areas

Top 5 PHC / UPHC areas — case profile and characteristics

PHC / UPHC	CASES	AGE RANGE	IMM. STATUS	KEY HSC SECTORS
Kaspa UPHC Urban PHC	7	1 mo – 18 yr	Completely Immunised (5), Unknown (2)	R.N.Palayam, Shariff Nagar, Chinnaallapuram, Kaspa
Sathuvachari UPHC Urban PHC	5	1 – 28 yr	Completely Immunised (3), Unknown (2)	Sathuvachari, Kagithapattarai
Latheri PHC Rural PHC	3	14 – 27 yr	All: Unknown status	Latheri sector
Pernambut UPHC Urban PHC	3	0 – 4 yr	Completely Immunised (3)	T.V.K.Nagar sector
Pallikonda PHC Rural PHC	3	1 – 7 yr	Completely Immunised (2), Unknown (1)	Pallikonda sector

Latheri PHC (adult cases, all Unknown status) and Kaspa UPHC (urban; highest case load) require prioritised surveillance response.

Key Findings

Critical epidemiological observations — Vellore District, Jan – Jul 2026

1. **Infants under 1 year bear the highest burden**
   28 cases (38.4%) are infants below 1 year of age — most are too young to receive MR-I at 9 months, making them inherently unprotected.

2. **March and June are high-transmission months**
   March (17 cases) and June (13 cases) recorded peak reporting, suggesting seasonal community transmission patterns warranting heightened surveillance.

3. **61.6% of cases had complete immunization — vaccine failure investigation required**
   45 of 73 cases had received MR vaccine as per schedule. This warrants cold chain review, vaccine efficacy assessment, and serological testing.

4. **Adults (16+) account for 23.3% of cases**
   17 adult cases (ages 18–50) detected, including cases with "Not Known" immunization history — indicating susceptibility gaps in older birth cohorts.

5. **Urban PHCs carry 65% of reported case load**
   Kaspa UPHC (7) and Sathuvachari UPHC (5) together account for 16.4% of all cases, pointing to urban transmission clusters.

Immunization Gap Analysis

Understanding why vaccinated children are still developing suspected measles

45 COMPLETELY IMMUNISED CASES — BREAKDOWN

28
Infants under 1 yr (vaccinated NA — too young)

17
Children / adults with documented vaccination

PROPORTION UNDER 9 MONTHS

38.4%
Cases ineligible for MR-I due to age — protection gap cannot be closed through routine immunization alone

Possible Reasons for Vaccine Failure

- Cold chain failure at point of vaccination
- Primary vaccine failure (5–10% with single dose)
- Documentation error — vaccine given but not recorded
- Waning immunity in children vaccinated years ago
- Misclassification of rubella as measles (clinical diagnosis)

RECOMMENDED ACTIONS

- Collect blood samples for IgM serology
- Review cold chain logs for affected HSC batches
- Cross-check immunization register vs. CIF records
- Notify State VPD Unit for virus genotyping
- Screen household contacts for outbreak containment

Recommendations

Priority public health actions across three domains

DOMAIN 01

Surveillance

- Strengthen PHC-level case detection and CIF submission within 24 hours of identification
- Collect blood samples for IgM serology from all suspected cases to confirm diagnosis
- Enhance weekly reporting from all 35 PHC / UPHC areas to detect clusters early
- Conduct contact tracing for each confirmed case, especially in urban clusters

DOMAIN 02

Immunization

- Review cold chain logs at Kaspa, Sathuvachari, and Lakshmipuram UPHCs — highest case load areas
- Conduct targeted MR-II follow-up campaigns for R.N.Palayam (+8) and Vallalar (+6) dropout sectors
- Ensure immunization of all children 9–12 months with MR-I as per schedule without delay
- Address "Not Known" immunization status (32.9%) through immunization register verification

DOMAIN 03

Outbreak Response

- Conduct rapid risk assessment for Kaspa and Sathuvachari UPHCs given sustained case reporting
- Declare outbreak response protocol if 3+ cases from same school or locality are confirmed
- Submit representative viral isolates for genotyping to identify circulating measles strains
- Coordinate with CMC and GVMCH for timely case notification and specimen referral

Conclusion

Summary and way forward for Vellore District VPD surveillance

73 suspected cases across 35 PHC / UPHC areas

Sustained transmission from January through July 2026, with March and June as peak months — both PHC and UPHC areas are affected.

High infant vulnerability requires structural response

38.4% cases are under 1 year — ineligible for routine MR-I. Antenatal care linkage and maternal immunity assessment should be considered.

Immunization system review is non-negotiable

With 61.6% of cases having complete immunization status, cold chain integrity and programme fidelity must be audited across all affected HSC sectors.

SNAPSHOT

Total cases	**73**
Under 1 year	**28 (38.4%)**
Immunized cases	**45 (61.6%)**
Tertiary-reported	**47 (64.4%)**
Peak month	**March (17 cases)**